CSCI 2001 - Object Oriented Programming (CS1)

Title: Bank Account Management System

Final Project Specification (**Total Points:** 80)

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Project Overview

In this project, students will implement a Bank Account Management System using Object-Oriented Programming (OOP) principles in Java. The system will support checking and savings accounts with deposit, withdrawal, interest calculation, and monthly processing functionalities.

This project is inspired by the programming assignment "9. BankAccount and SavingsAccount Classes" on Page 698 (in printed text book), Chapter 10. Inheritance.

However, the assignment outlined here has some additional classes as well. While the specificaiton (Programming Assignment 9) in the textook outlines two classes, your final project has 6 different classes. Please review the specification of these classes here.

Also note that I have provided test code for many of earlier assignments in this course. As part of the final project, you will be writing your own tester that displays the passing test cases, failing test cases, and the final score. I reserve the right to use my own tester or use one student's tester against another student's submission. As long as the names of the functions and classes are honored, the testers can be interchanged.

Project Requirements

1. BankAccount Interface (5 points)

- Define an interface BankAccountInterface that includes:
    - void deposit(double amount)
    - void withdraw(double amount)
    - void calculateInterest()

- void monthlyProcess()

2. Abstract Class: BankAccount (10 points)

- Implement an abstract class BankAccount that:
  - Implements BankAccountInterface.
  - Includes attributes:
    - double balance
    - int numDeposits
    - int numWithdrawals
    - double annualInterestRate
    - double monthlyServiceCharge
  - Implements common methods such as deposit, withdraw, calculateInterest, and monthlyProcess.

3. SavingsAccount Class (10 points)

- Inherit from BankAccount.
- Implement additional features:
  - An isActive flag (accounts with balance < 25 become inactive).
  - Withdrawals should only be allowed when active.
  - Monthly processing applies a $1 service charge for each withdrawal beyond 4.

4. CheckingAccount Class (10 points)

- Inherit from BankAccount.
- Implement additional features:
  - A $0.50 withdrawal fee.
  - A minimum balance of $50.
  - A $10 monthly fee if balance falls below $50 at the end of the month.

5. CS1Bank Class (10 points)

- Maintain a list of Savings and Checking accounts.

- Implement methods to:
    - Add new accounts.
    - Process monthly updates for all accounts.

6. BankAccountTester Class (25 points)

- Implement a test class that:
    - Tests CheckingAccount functionality (30 points total in grading, partial points awarded per test case).
    - Tests SavingsAccount functionality (30 points total in grading, partial points awarded per test case).
    - Tests CS1Bank functionalities (20 points total in grading, partial points awarded per test case).
    - Prints results showing which test cases pass or fail.
    - Displays a final score out of 80.

7. Output file, Coding Standards and Documentation (10 points)

- You must submit the output from your tester
- Code must follow Java conventions:
    - Use meaningful variable/method names.
    - Proper indentation and formatting.
    - Add inline comments explaining logic.
    - Include Javadoc comments for classes and methods.

Grading Breakdown

Task	**Points**
BankAccount Interface	5
Abstract Class: BankAccount	10
SavingsAccount Class	10

Task	Points
CheckingAccount Class	10
CS1Bank Class	10
BankAccountTester Class	25
Output of your tester class	10
Total	**80**

Note: Partial credit will be awarded for incomplete implementations that demonstrate effort and correctness.

What to submit?

1. A zip file of all the java files
2. Output from your tester class as a text file.